



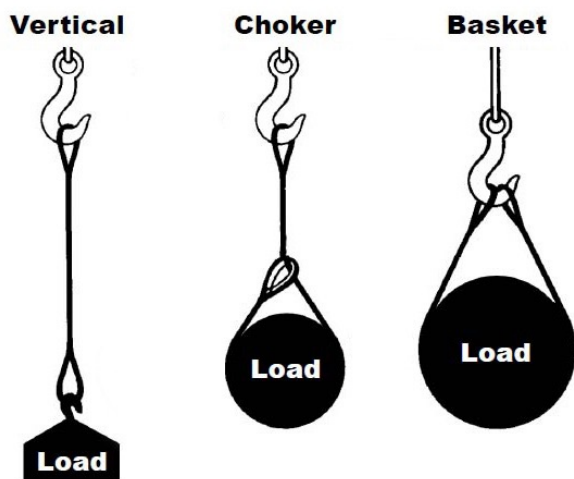
WEB SLINGS

CoreLifting.com



General Information

Basic Hitches



VERTICAL, or straight, attachment is simply using a sling to connect a lifting hook to a load. Full rated lifting capacity of the sling may be utilized but must not be exceeded. A tagline should be used to prevent load rotation, which may damage a sling. When two or more slings are attached to the same lifting hook, the total hitch becomes, in effect, a lifting bridle, and the load is distributed equally among the individual slings.

CHOKER hitches reduce lifting capability of a sling since this method of rigging affects ability of the wire rope components to adjust during the lift. A choker is used when the load will not be seriously damaged by the sling body — or the sling damaged by the load, and when the lift requires the sling to snug up against the load.

The diameter of the bend where the sling contacts the load should keep the point of choke against the sling **BODY** — never against a splice or the base of the eye. When a choke is used at an angle of less than 120 degrees (see next page), the sling-rated capacity must be adjusted downward.

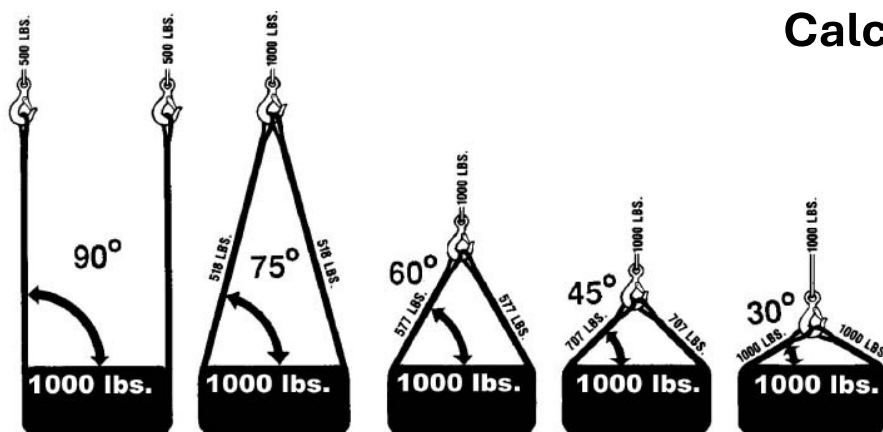
A choker hitch should be pulled tight before a lift is made — **NOT PULLED DOWN DURING THE LIFT**. It is also dangerous to use only one choker hitch to lift a load which might shift or slide out of the choke.

BASKET hitches distribute a load equally between the two legs of a sling ... within limitations described above.

General Information

Load Calculating

Calculating the Load on Each Leg of a Sling



As the included angle between the legs of a sling decreases, the load on each leg increases. The effect is the same whether a single sling is used as a basket or two slings are used with each in a straight pull, as with a 2-legged bridle.

Anytime pull is exerted at an angle on a leg—or legs—of a sling, the load per leg can be determined by using the data in the table above. Proceed as follows to calculate this load—and determine the rated capacity required of the sling, or slings, needed for a lift.

1. First, divide the total load to be lifted by the number of legs to be used. This provides the load per leg if the lift were being made with all legs being vertically.
2. Determine the angle between the legs of the sling and the horizontal.
3. Then MULTIPLY the load per leg (as computed above) by the Load Factor for the leg angle being used (from the table at the bottom) – to compute the ACTUAL LOAD on each leg for this lift and angle. THE ACTUAL LOAD MUST NOT EXCEED THE RATED SLING CAPACITY.

Thus, in the above drawing (sling angle at 60°): $1000 \div 2 = 500$ (Load Per Leg if a vertical lift) $500 \times 1.154 = 577$ lbs. – ACTUAL LOAD on each leg at the 60° included angle being used.

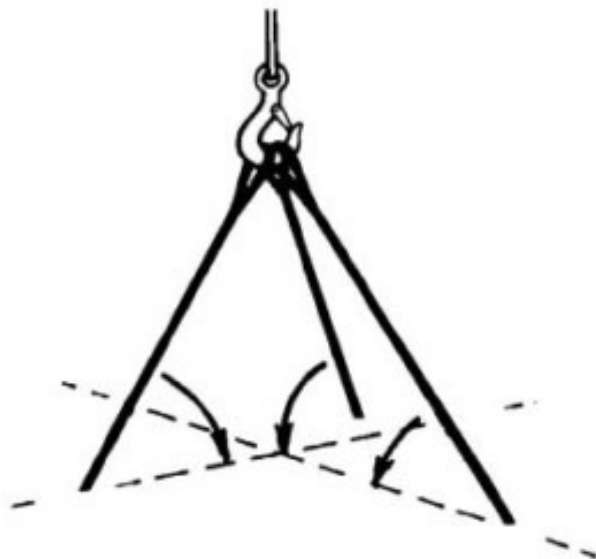
In the above drawing (sling angle of 45°): $1000 \div 2 = 500$ (Load Per Leg if a vertical lift) $500 \times 1.414 = 707$ lbs. = ACTUAL LOAD on each leg at the 45° horizontal angle being used.

General Information

Bridle Legs – Angle Effects

Angles of Bridles

The horizontal angle of bridles with 3 or more legs is measured the same as the horizontal sling angle of 2-legged hitches. In this case, where a bridle designed with different leg lengths results in horizontal angles, the leg with the smallest horizontal angle will carry the greatest load. Therefore, the smallest horizontal angle is used in calculating actual leg load and evaluating the rated capacity of the sling proposed.



Leg Angle (Degrees)	Load Factor
90	1.000
85	1.003
80	1.015
75	1.035
70	1.064
65	1.103
60	1.154
55	1.220
50	1.305
45	1.414
40	1.555
35	1.743
30	2.000

General Information

Web Sling Configurations

Web slings are pretty similar in configuration to other sling types, like wire rope slings.

Web slings come in several different configurations, including:

- Single or multi-leg bridle assemblies
- Eyes formed on both ends of the sling
- An endless or infinite loop configuration (similar to a synthetic roundsling)
- With fittings on both ends of the sling.

Hardware manufacturers make special hooks for web slings, and instead of your traditional eye on a hook, they actually make a flat surface on the hook. This way, the web sling will sit properly against the hook without getting cinched up, which allows you to get the maximum lifting strength.



EYE & EYE



ENDLESS

General Information

Web Sling Types

Type I Triangle/Choker - (TC)

Web sling made with a triangle fitting on one end and slotted triangle choker fitting on the other end. It can be used in a vertical, basket or choker hitch.



Type II Triangle x Triangle - (TT)

Web sling made with a triangle fitting on both ends. It can be used in a vertical or basket hitch only.



Type III Eye x Eye - (EE)

Web sling made with a flat loop eye on each end with loop eye opening on same plane as sling body. This type of sling is sometimes called a flat eye x eye or double eye sling.



Type IV Reversed Eye - (RE)

Web sling made with both loop eyes formed as in Type III, except that the loop eyes are turned to form a loop eye, which is at a right angle to the plane of the sling body. This type of sling is commonly referred to as a twisted eye sling.



Type V Endless Type - (EN)

Endless web sling, sometimes referred to as a grommet. It is a continuous loop formed by joining the ends of a piece of webbing together with a load bearing splice.



General Information

Measuring Sling Length

How to accurately measure a web sling.

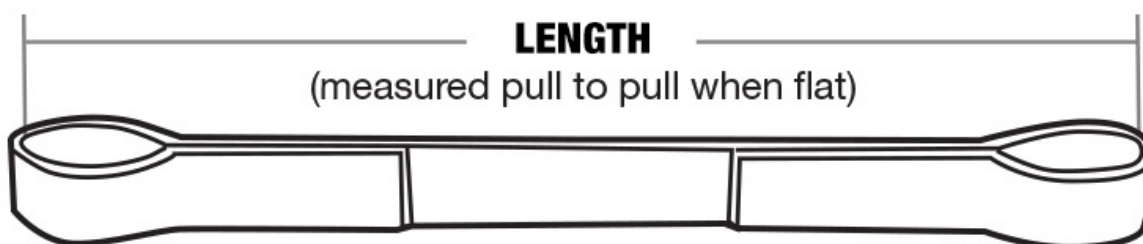
Synthetic web slings can come in various lengths. Many fabricators have standardized product offerings, including incremental increases in length ranging anywhere from 2' to 20'. However, it's also common for fabricators to offer customized sling lengths, so you can get a web sling designed to your application's exact specifications.

When determining sling length, measure from one load-bearing point to the opposite load-bearing point.

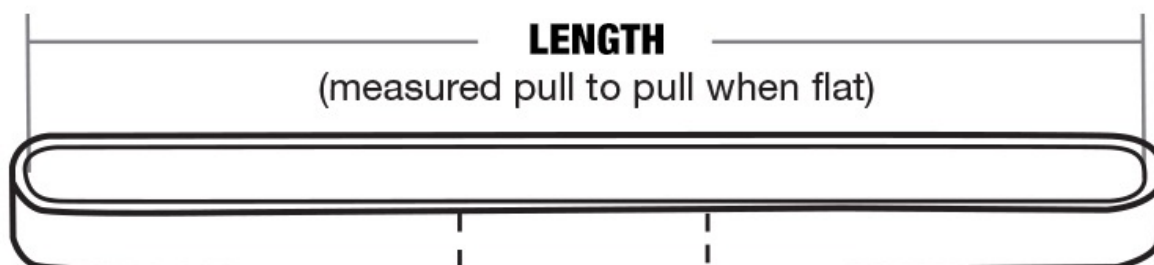
An eye and eye sling length is measured from the inside edge of the eyes.

Multi-leg sling assemblies, or single-leg slings with fittings, are measured from the load-bearing point of rings, hooks, or fittings.

EYE AND EYE



ENDLESS





General Information

Polyester vs Nylon

Polyester Slings and Nylon Slings

For many years, most of the world outside of the United States used Polyester slings, while the U.S. used Nylon. Though nearly identical in strength, there are differences in how each reacts to loading and they're interaction with chemicals. In the past 10 years, the U.S. market has largely migrated to Polyester, and this is the primary webbing fiber used in Core-Tex web slings.

Polyester Web Slings

- Approximately 3% stretch at rated capacity for more load control.
- Polyester is a slightly softer fiber and less likely to scratch surfaces of what is being lifted.
- Absorbs less liquid than Nylon.
- Is non-conductive.
- Resistant to acidic environments and bleaching agents.
- Is less costly than Nylon.
- Not recommended for alkaline environments including aldehydes, ethers, and strong alkalis.
- Cannot be used in temperatures greater than 194F or below -40F.

Nylon Web Slings

- 8-10% stretch at rated capacity. This helps absorb shock loads, but this must be accounted for, especially in low headroom situations.
- Unaffected by grease and oil.
- Resistant to alkalis such as aldehydes, ethers and strong alkalis.
- Not recommended for acidic environments or exposure to bleaching agents.
- Absorbs liquid which can add to or cause more stretch under load.
- Will conduct electricity.
- Cannot be used in temperatures greater than 194F or below -40F.

General Information

Plies

Number of Plies

The ply number, or number of plies, refers to the layers of webbing that make up the body of the sling. A web sling is generally available in 1, 2, 3, or 4-ply configurations. Adding more plies to the sling provides added strength to the body of the sling.

Lifting capacity is going to determine how many plies are needed to execute your lifts safely. More times than not, when you have higher-capacity lifts, you're going to need more plies on your web slings.

For example, if the breaking strength of a 1-ply web sling = X, then...

- The breaking strength of an equivalent 2-ply web sling = 2X
- The breaking strength of an equivalent 3-ply web sling = 3X
- The breaking strength of an equivalent 4-ply web sling = 4X





General Information

Width

Web Sling Width

With a wider sling body, the load is distributed over a larger sling surface area. Wider width slings provide greater protection to delicate or finished surfaces, as well as better load balance and control for large or heavy loads.

Typical standard sling lengths can range anywhere from 1" to 12" in width. Larger sizes are available as special-orders, or in a special wide lift sling configuration.

If you don't want to use a three-ply web sling, you may want to increase the width of your synthetic material for better load control.

A lot of times, if you're using your sling in a basket hitch configuration, that additional width is going to help keep a load balanced more than using a thinner sling. With a thinner sling, the load might move if you're just using one. Determining what you need to execute your lifts is key to finding the right rigging gear.

How width affects strength:

Sling webbing, both polyester and nylon, has a typical breaking strength of 9,800 lbs per inch. So, 1 inch webbing has a breaking strength of 9,800 lbs. 2 inch wide webbing has double that or 19,600 lbs of breaking strength.

The number of plies and width of webbing combine to determine breaking strength of the sling. So, a 2 inch wide web sling that has 2 plies has a breaking strength in vertical configuration of $2 \times 9,800 \text{ lbs} \times 2 \text{ plies} = 39,200 \text{ lbs}$ ($2 \times 9,800 = 19,600 \text{ lbs} \times 2 = 39,200 \text{ lbs}$).

General Information

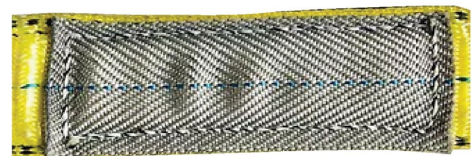
Webbing Protection

Webbing Can Wear

Sling webbing is a fiber product vulnerable to abrasion. To guard against this, obvious wear points where the sling comes into contact with end fittings such as shackles, master links, hooks, etc., more abrasion resistant fiber webbing products such as Cordura can be sewn into the eyes of the slings.

When the sling is used in a basket configuration, Cordura type products can be wrapped over the area most vulnerable to wear and sewn onto the sling surface.

Other options are wear-sleeves that can be slid over the sling's surface.



Web Slings

Type I & Type II

Loop Eye Table		
Sling Web Width (in.)	Recommended Min Loop Eye Length (in.)	Suggested Loop Eye Width (in.)
1	6	1
2	8	2
3	8	1.5
4	10	1.5
5	12	1.75
6	14	2
8	18	3
10	22	4
12	26	5

Plies	Vertical lbs.	Choker lbs.	Basket 90 deg. lbs.	Web Width in.
1	3,200	2,560	6,400	1
1	6,200	4,960	12,400	2
1	9,400	7,520	18,800	3
1	12,400	9,920	24,800	4
1	18,600	14,880	37,200	6
2	6,200	4,960	12,400	1
2	12,400	9,920	24,800	2
2	17,600	14,080	35,200	3
2	22,000	17,600	44,000	4
2	33,000	26,400	66,000	6

Tapering: Flat, Twisted Eye and Endless Slings are tapered at 3" and wider unless specified differently when ordering.

*** Please Note: Polyester webbing is not available over 10" wide.**

Three and four-ply slings are available.

**** TT can not be used in a choker hitch.**

Type I Triangle / Choker Fittings (TC) Web Slings

- These types of slings are configured with a triangle and choker fitting on either end. These types of slings are typically used in a choker hitch and can be used in vertical and basket hitches as well.



Type II Triangle / Triangle Fittings (TT) Web Slings

- This type of sling is configured with a triangle fitting on each end. These types of slings are normally used in a basket hitch but can also be used in a vertical hitch. They cannot be used as a choker.



TT1 & TT2 Specifications

- Web thickness of "Heavy-Duty" web is approximately 3/16".
- Eye length of Flat Eye and Twisted Eye varies with sling width and number of webbing piles.

Web Slings

Type III & Type IV



Plies	Vertical lbs.	Choker lbs.	Basket 90 deg. lbs.	Web Width in.	Eye Length in.	Eye Width in.
1	1,600	1,280	3,200	1	9	1
1	3,100	2,480	6,200	2	9	2
1	4,700	3,760	9,400	3	12	1.5
1	6,200	4,960	12,400	4	14	2
1	9,300	7,440	18,600	6	18	2
1	11,800	9,440	23,600	8	22	3
1	14,700	11,760	29,400	10	28	3.5
1	17,600	14,080	35,200	12	30	4
2	3,100	2,480	6,200	1	9	1
2	6,200	4,960	12,400	2	9	2
2	8,800	7,040	17,600	3	12	1.5
2	11,000	8,800	22,000	4	14	2
2	16,500	13,200	33,000	6	18	2
2	22,700	18,160	45,400	8	22	3
2	28,400	22,720	56,800	10	28	3.5
2	34,100	27,280	68,200	12	30	4
3	4,700	3,760	9,400	1	12	1
3	9,300	7,440	18,600	2	12	2
3	13,500	10,800	27,000	3	14	1.5
3	17,200	13,760	34,400	4	18	2
3	24,400	19,520	48,800	6	24	3
3	30,700	24,560	61,400	8	26	4
3	36,000	28,800	72,000	10	30	5
3	40,300	32,240	80,600	12	42	6
4	5,500	4,400	11,000	1	12	1
4	11,000	8,800	22,000	2	12	2
4	16,400	13,120	32,800	3	14	1.5
4	20,400	16,320	40,800	4	18	2
4	30,800	24,640	61,600	6	26	3
4	41,000	32,800	82,000	8	28	4
4	48,000	38,400	96,000	10	32	5
4	60,000	48,000	120,000	12	44	6

Type III Eye x Eye Flat Eyes Web Slings

- These types of slings have flat eyes and can be made in widths 1" thru 12" and 1, 2, 3 or 4 plies.



Type IV Eye x Eye Reverse Eyes Web Slings

- These types of slings have reverse eyes and can be made in widths 1" thru 12" and 1, 2, 3 or 4 plies.



Type III & IV Specifications

- Flat and reverse eyes are tapered beginning at 3" and wider unless specified when ordering.



Web Slings Type V



Plies	Vertical lbs.	Choker lbs.	Basket 90 deg. lbs.	Web Width in.
1	3,200	2,560	6,400	1
1	6,200	4,960	12,400	2
1	9,400	7,520	18,800	3
1	12,400	9,920	24,800	4
1	18,600	14,880	37,200	6
1	21,200	16,960	42,400	8
1	26,500	21,200	53,000	10
1	31,800	25,440	63,600	12
2	6,200	4,960	12,400	1
2	12,400	9,920	24,800	2
2	17,600	14,080	35,200	3
2	22,000	17,600	44,000	4
2	33,000	26,400	66,000	6
2	42,300	33,840	84,600	8
2	52,900	42,320	105,800	10
2	63,500	50,800	127,000	12
3	9,600	7,680	19,000	1
3	17,700	14,160	35,400	2
3	25,600	20,480	51,200	3
3	34,600	27,680	69,200	4
3	48,900	39,120	97,800	6
3	61,400	49,120	122,800	8
3	72,000	57,600	144,000	10
3	80,600	64,480	161,200	12
4	11,000	8,800	22,000	1
4	22,000	17,600	44,000	2
4	32,900	26,320	65,800	3
4	40,800	32,640	81,600	4
4	61,200	48,960	122,400	6
4	81,900	65,520	163,800	8
4	96,000	76,800	192,000	10
4	107,000	85,600	214,000	12

Type V Endless Web Slings

- These types of slings can be made in widths 1" thru 12" and 1, 2, 3 or 4 plies.



Web Slings

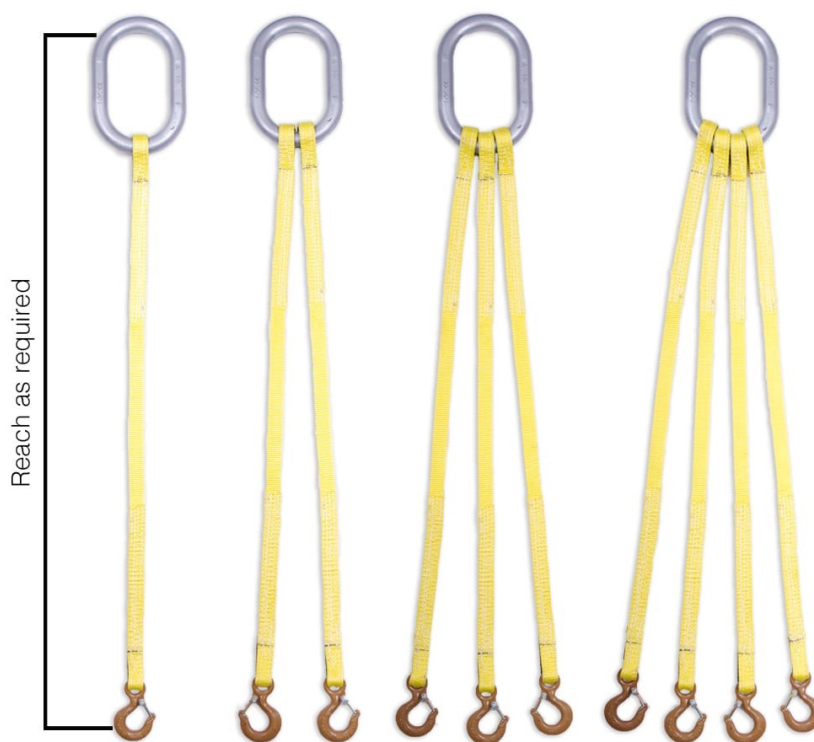
Multi-Leg Web Sling Bridles



Sling Type	Webbing Width (in.)	Number of Plies	Number of Legs	Rated Capacities (lbs.)				Master Link Size (in.)	Alloy Hooks Capacity * (ton)
				Vertical	60°	45°	30°		
EE1-801	1	1	1	1600	–	–	–	1/2 x 2-1/2 x 5	1
EE1-801	1	1	2		2700	2200	1600	1/2 x 2-1/2 x 5	1
EE1-801	1	1	3		4100	3300	2400	3/4 x 2-3/4 x 5-1/2	1
EE1-801	1	1	4		5500	4500	3200	1 x 3-1/2 x 7	1
EE2-801	1	2	1	3000	–	–	–	1/2 x 2-1/2 x 5	1-1/2
EE2-801	1	2	2		5100	4200	3000	3/4 x 2-3/4 x 5-1/2	1-1/2
EE2-801	1	2	3		7700	6300	4500	3/4 x 2-3/4 x 5-1/2	1-1/2
EE2-801	1	2	4		10300	8400	6000	1 x 3-1/2 x 7	1-1/2
EE1-802	2	1	1	3000	–	–	–	1/2 x 2-1/2 x 5	1-1/2
EE1-802	2	1	2		5100	4200	3000	3/4 x 2-3/4 x 5-1/2	1-1/2
EE1-802	2	1	3		7700	6300	4500	3/4 x 2-3/4 x 5-1/2	1-1/2
EE1-802	2	1	4		10300	8400	6000	1 x 3-1/2 x 7	1-1/2
EE2-802	2	2	1	6000	–	–	–	3/4 x 2-3/4 x 5-1/2	3
EE2-802	2	2	2		10300	8400	6000	1 x 3-1/2 x 7	3
EE2-802	2	2	3		15500	12700	9000	1 x 3-1/2 x 7	3
EE2-802	2	2	4		20700	16900	12000	1-1/4 x 4-3/8 x 8-3/4	3

- SOE = Single leg with oblong master link and eye on bottom
- SOS = Single leg with oblong master link and sling hook on bottom

There are many options for end fittings. Please see Core Synthetic Sling Hardware Catalog for specifications.



WEB SLING SAFETY BULLETIN



WARNING

Lifting & lifting systems require the use of several tools, variables & devices, and polyester web sling(s) are only one part of that system. It is critical that you read and understand this bulletin prior to using polyester web slings and the failure to do so, may result in severe **INJURY** or **DEATH** due to sling failure and/or loss of the load.

The 6 Areas of Concentration

1. TRAINING & KNOWLEDGE

2. GOOD INSPECTION PRACTICES

3. SLING PROTECTION

4. NEVER EXCEED RATED CAPACITY

5. STAY CLEAR OF LOAD

6. SLING STORAGE

1. TRAINING & KNOWLEDGE

Knowledge is acquired through training. Sling users must be knowledgeable in the safe and proper use of polyester web slings to prevent **injury** and **death** to you and others.

In addition to the information found in this bulletin, polyester web sling users should also read and understand the standards and guidance provided in:

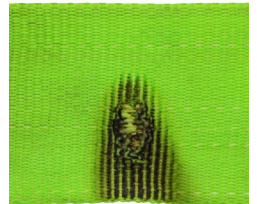
- ASME B30.9-5 Synthetic Webbing Slings: Selection, Use & Maintenance
- OSHA 29 CFR 1910.184: Slings
- OSHA Guidance On Safe Sling Use

2. GOOD INSPECTION PRACTICES

Because polyester web slings are made from synthetic fibers, they are vulnerable to damage in a variety of ways. Before using a sling to make a lift, it is necessary to inspect the sling for damage prior to use to prevent **injury** or **death** to you and others. Inspection requires you to **LOOK & FEEL** over the entire length of the sling on both sides.

During inspection, should you identify any of the following conditions, the sling is to be **removed from service** in accordance with ASME B30.9-5.9.4.

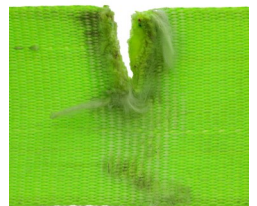
Melting or charring of any part of the sling



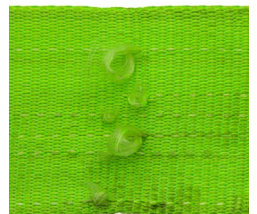
Holes/Punctures



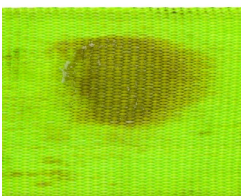
Cuts or Tears



Snags

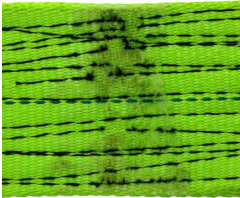


Acid or Alkali Burns

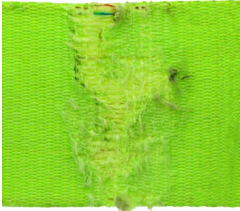


Knots

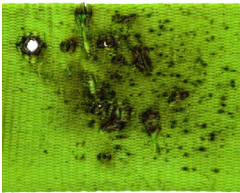




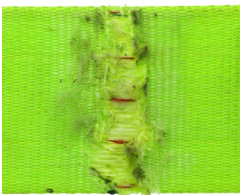
Broken or Worn Stitches



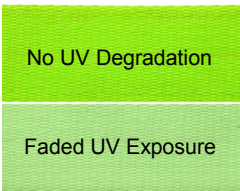
Excessive Abrasive Wear



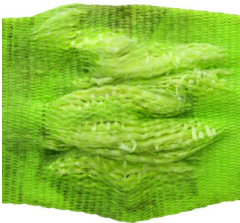
Weld Spatter



Exposed Red Core Yarns



UV Degradation



Crushed Webbing



Embedded Materials

Other removal criteria:

- Slings with fittings that are pitted, corroded, cracked, bent, twisted, gouged or broken.
- For slings with hooks sewn-in, apply removal criteria for hooks as stated in ASME B30.10.
- For other sewn-in rigging hardware, apply removal criteria as stated ASME B30.26.
- Other conditions, including visible damage, that cause doubt as to the continued use of the sling.

Polyester web slings should be inspected as follows:

Initially—When slings are received by a qualified person to ensure there is no damage and that the sling meets the requirements for its intended use.

Frequently—Polyester web slings should be inspected before using each shift or each day.

Periodically—Polyester web slings are to be inspected by a qualified person, other than the person who conducts the frequent inspections. Periodic inspections are to be done annually for normal service and monthly to quarterly for severe service. Inspection intervals are not to exceed one year. For periodic inspections, both WSTDA WS-1 and ASME B30.9 require that a written record of the inspections be maintained.

3. SLING PROTECTION

To protect polyester web slings from both environmental and physical damage, avoid the following:

- Continuous or excessive exposure to sunlight.
- Contact with dirt or gritty type substances.
- Dragging slings on the floor, ground or rough surfaces.
- Jerking or pulling slings out from under loads.
- Twisting or tying knots in slings.
- Allowing slings to contact acids or alkalis.
- Exposing slings to temperatures above 194°F or below -40°F.
- Allowing weld spatter to land on slings.
- Allowing sling to sit on the tip of a hook instead of the base of the hook's bowl.
- Using rigging hardware such as hooks or shackles with rough edges.
- Running or driving over slings with vehicles or mobile equipment.

To help prevent damage from sharp edges, corners and other abrasive surfaces, protective devices such as wear pads, sleeves, corner protectors and body wraps should be used. It may also be necessary to test samples to ensure protection method chosen is effective prior to lifting actual load to be lifted.

4. NEVER EXCEED RATED CAPACITY

To avoid exceeding the rated capacity of a polyester web sling, a trained and qualified person must first know the weight of the load to be lifted and its center of gravity to determine if the rigging system will sufficiently retain and control the lifted load.

Also, determine whether the sling will be used in a vertical hitch, choker hitch or vertical basket hitch. See table below for description and illustration.

Vertical

One end is placed on the hook and the other end is attached directly to the load. A tagline should be used to prevent load rotation.

Choker

Sling passes thru one end & around the load while other end is placed on hook. Rated capacity is normally 80% of vertical hitch. Choke point should always be on sling body, not sling eye, fitting, splice or tag.

Vertical Basket

Sling cradles load while ends are attached overhead. Rated capacity is two times that of vertical hitch.

It is also very important that a qualified person determine the angle that the slings will be placed in for the intended lift. This may even be a challenge for a qualified person. But for someone unqualified and unaware of the effects that sling angles can have on the rated capacity of the slings, the lack of this knowledge or inability to apply the correct sling reduction factors can lead to injury or death.

WARNING: ANGLES MATTER!

For Vertical & Basket hitches, sling angles less than 90 degrees **reduce the effective rated capacity of the sling.** See table below.

Angle	Sling Reduction Factor
90°	1.000
85°	0.996
80°	0.985
75°	0.966
70°	0.940
65°	0.906
60°	0.866
55°	0.819
50°	0.766
45°	0.707
40°	0.643
35°	0.574
30°	0.500

Vertical & Basket Hitches

To calculate the effective capacity of the sling, determine the angle of the sling and multiply the sling reduction factor for the determined angle times the rated Vertical or Basket capacity of the sling.

Angle of Choke		Sling Reduction Factor
= or >	<	
120°	180°	1.00
105°	120°	0.82
90°	105°	0.71
60°	90°	0.58
0°	60°	0.50

For Choker hitches, to calculate the effective capacity of the sling, determine the angle of choke and multiply the sling reduction factor for the determined angle of choke times the rated Choker capacity of the sling.

5. STAY CLEAR OF LOAD

- Never stand in the area directly beneath or near the load being lifted.
- Never stand on load being lifted.
- Never allow any part of your body to be placed between the sling and the load being lifted or between the sling and the lifting hook or rigging fitting.
- Be sure all personnel in the area of the lift are alert to the potential for the sling to become snagged during the lift.
- Never use a web sling to pull on objects in a constrained or snagged condition.

6. SLING STORAGE

- Polyester web slings should always be stored in dry, dark and cool places when not being used.
- Polyester web slings should always be stored away from exposure to chemicals, high heat sources, metal turnings from machining, grinding dust, splinters, etc.
- Always be sure polyester web slings are stored in a place free from dirt, oils, grime and other such materials.
- Clean only with mild soap and water. If washed, be sure polyester web sling is 100% dry before being stored. Never machine wash as it will result in loss in strength.

There are many factors that should be considered when using web slings to lift loads. Some of these factors include:

- | | |
|---------------------|------------------------|
| ▪ Wind | ▪ Weather |
| ▪ Visibility | ▪ Ground Stability |
| ▪ Air Temperature | ▪ Weight |
| ▪ Center of Gravity | ▪ Dimensions |
| ▪ Clear Load Path | ▪ Structural Stability |
| ▪ Sling Selection | ▪ Load Control |
| ▪ Lift Point | ▪ Sling Capacity |
| ▪ Signals | ▪ Planning |

The information in this bulletin does not contain ALL of the information required to use web slings safely. Other recommended sources of information include:

- OSHA 29CFR 1910.184
- Web Sling Tie Down Association (WSTDA) "WS-1" Recommended Standard for Synthetic Web Slings
- ASME B30.9 Sling Standard



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